

Generation Interconnection System Impact Study Report

For Queue Project Q19-1234

Dpc Elroy 138 kV
December 23, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1234. The Interconnection Customer proposes a 193.7 MW battery energy storage facility interconnecting at the Dpc Elroy 138 kV substation in the Virginia area. The request was received on June 27, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 394 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 9 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$5,652,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Walthall – Haggin Substation 138 kV	N Tower Tap – Haggin 345 kV line	111.4	563	27.5
Cedar Bayou Plant Substation – Centralia 138 kV line	Potosi – Tyler Substation 138 kV	111.4	208	7.4
Deltaport – Potosi Substation 138 kV line	Haggin – Jim Hill 138 kV	105.5	301	7.3
Tyler – Potosi 138 kV line	Jim Hill – Haggin 138 kV	103.8	332	7.3
Yellow Crk – Walhall 138 kV	N Tower Tap – Unknown Substation 138 kV	103.4	298	10.2
Geneva Border City – Keokuk 138 kV line	Dpc Elroy – N Tower Tap 138 kV line	102.4	294	36.1
N Tower Tap – Dpc Elroy 138 kV	Mccartney – North Tower Tap 345 kV L	102.3	561	66.6
Jim Hill – Haggin 138 kV	Potosi – Tyler 138 kV	100.8	324	7.4
Chambersburg 500/345kV Transformer	East Devon – Thiele Kaolin Reedy Creek 345 kV	100.3	1,030	5.3

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Walthall – Haggin 138 kV	Rebuild line with higher-capacity conductor	\$4,548,000	14% (\$635,000)
Cedar Bayou Plant Substation – Centralia 138 kV line	Rebuild line with higher-capacity conductor	\$988,000	18% (\$180,000)
Potosi – Deltaport Substation 138 kV	Reconductor line	\$2,800,000	17% (\$467,000)
Potosi – Tyler 138 kV	Reconductor line	\$190,000	17% (\$32,000)
Yellow Crk – Walhall 138 kV	Reconductor line	\$1,026,000	25% (\$253,000)
Geneva Border City – Keokuk 138 kV Line	Reconductor line	\$3,940,000	61% (\$2,397,000)
N Tower Tap – Dpc Elroy 138 kV	Reconductor line	\$1,424,000	61% (\$864,000)
Jim Hill – Haggin 138 kV	Reconductor line	\$212,000	18% (\$39,000)
Chambersburg 500/345kV XFMR	Replace transformer with higher-rated unit	\$30,426,000	3% (\$785,000)
	Total allocated to project		\$5,652,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.